

INTRO FINITE ELEMENT – SPRING 2022

Project #2

Student Names:

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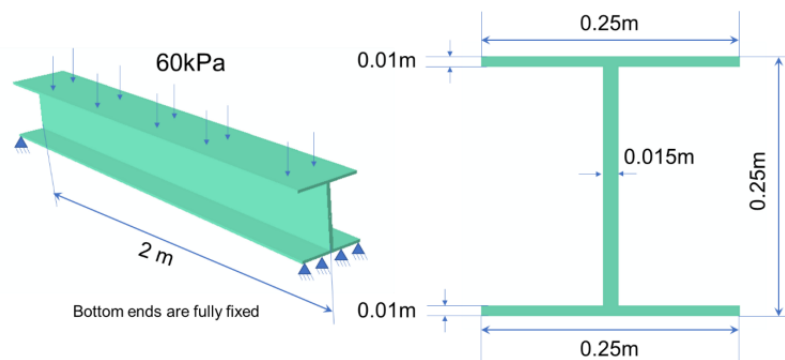


Figure 1: I-Beam Parameters

Introduction:

Project 2 dealt with a deeper analysis of mesh element types and their effect on model results. A general case with an I-beam was used for this analysis. The upper flange had an applied pressure of 60kPa. Three distinct element types were modeled and analyzed: The 2D beam element, the shell element, and the solid element. In addition to the mesh elements, the effects of mesh density on the results were also discussed. An academic license for ABAQUS was used with the 2021 Abaqus version to simulate the I-beam tests.

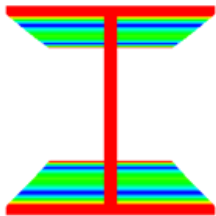
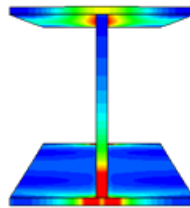
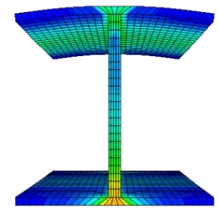
Abstract:

After sorting through the vast amounts of data collected, the three mesh element types' pros and cons were illuminated. The beam element, although the simplest mesh element type, still has a wide variety of uses in industry. Some of the positives of this element type include low computational expense and the simplicity of the element type. However, the negatives outweigh the positives when you consider that modern computers usually have enough processing power to tackle more complicated calculations in a reasonable amount of time. Although the element type was very computationally inexpensive, it was inaccurate when trying to account for any displacement in the 3rd dimension, as it is a 2D mesh element type. However, its stress was quite comparable to the other element types. Overall, the Beam element type is great for a quick simple 2D understanding of a structure's performance under a load, but more computationally intensive methods are needed to get an accurate picture.

The shell element provided the best combination of computational efficiency and material deformation accuracy. It was more computationally efficient than the Solid element type, however, it also had many more variables to edit than the 2D beam element had, thus increasing accuracy. This element type was also able to account for deformation in the 3rd dimension but was unable to account for shell thickness varying due to loads. However, for all its praises, this element type has some drawbacks; it is not as accurate as the solid element type, it is slower than the 2D beam type, and is more complicated to mesh and create for novice users.

The solid element gave the best results of all beam element types. Its best aspect is that it can be used to model very complex geometry, whereas the 2D beam element could only model 2D objects, and the shell element type was much more complicated to model complex

geometries because it used 2D planes for all its elements. Another benefit that the solid beam element provides is a high degree of accuracy. It can account for loads in all directions and is thus extremely versatile. It can also show stress more accurately, whereas other element types would not be able to show the minute details that the solid element can provide. Another benefit of the solid element is the fact that it has a multitude of mesh options. Not only can a structured mesh be used, but tet and hex meshes can also be used among others which are too numerous to be listed in this document. The main downside is that it was also the most computationally expensive element type of the three.

**Figure 2: Beam Element****Figure 3: Shell Element****Figure 4: Solid Element**

Results from each element type. This highlights the difference between each element type.

Problem Parameters and Solution:

Using ABAQUS, a Finite Element Modeling Software, an I-beam was thus created with an overall height of 0.25m, flange thickness of 0.01m, flange width of 0.25m, web thickness of 0.015m, and an overall length of 2 meters. The upper and lower flange dimensions were identical. The I-beam material consisted of stainless steel ($E=200\text{GPa}$ and $\nu=0.3$). Supports were placed at both ends of the bottom flange and a pressure with a magnitude of 60kPa was placed on the top flange. We then compared 3 different mesh elements each with a multitude of mesh densities.

Solution Approach:

For the “2D Beam Element,” we created a 2D Wire part. We drew a line 2 m long and created an I-Beam profile with the required specifications. We applied a section assignment and placed the loads and BCs, then applied a mesh and ran the job file. After reviewing the results of the job file, we analyzed the results and noted the max displacement magnitude and the max stress. For the shell element, we created a simple 3D shell extrusion. We drew a simple I shape for our cross-section sketch. We then extruded it to a length of 2m. Next, we created 2 sections and assigned them accordingly. Finally, we applied the loads and BCs and ran the job. We analyzed the results and noted the max displacement magnitude and the max stress. For the Solid Element, we created a section sketch of the beam profile, extruded it to 2 meters, assigned a section, placed loads and BCs, and ran the job file. We noted that the results for this beam element differed drastically from the other 2 element types. We noted that the displacement differed from the other element types by an order of magnitude.

Results/Analysis:

After an extensive group effort, we managed to sift through all the data collected throughout the 2 weeks spent on this project. After using a wide range of tools for data processing and a few group discussions, we decided by majority vote on the best beam element type. After taking into consideration all currently known factors for the beam element, we determined that it should only be used in situations where the computational expense is very limited, as it is the least computationally expensive mesh element type. However, when available, the shell or solid element type is highly recommended, as the drawbacks of using the beam element type are numerous. The drawbacks can include but are not limited to very low mesh density, two-dimensional loading, nuances not being highlighted such as flange deteriorations from loading, and fewer options for BCs.

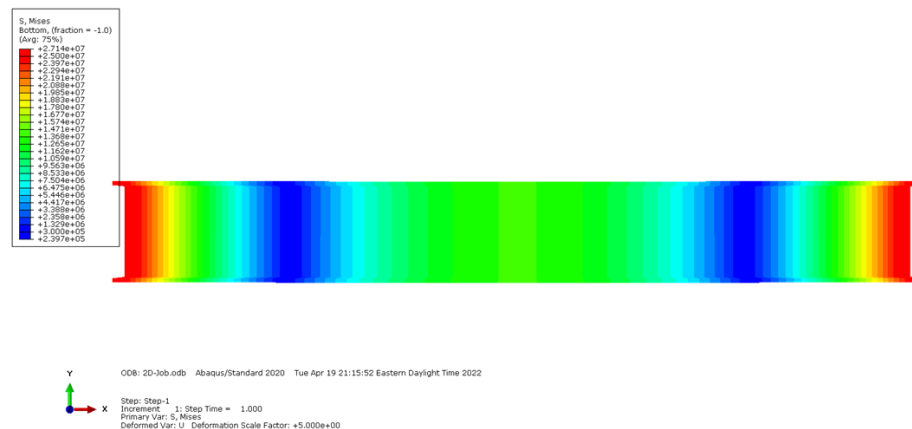


Figure 5: Beam Element

After taking a detailed look at the shell element, we concluded that it combines the best features of the other 2 element types, without as many drawbacks; however, for best results, the solid element type should be used. Firstly, the shell element type is much less computationally expensive than the solid beam element type, yet not much more computationally expensive than the 2D beam element. Secondly, it can account for flange displacement and can be used to diagnose where the I-beam should be reinforced, whereas the beam element cannot identify where the beam profile is weakest. The major drawback is that because it is also a 2D simulation type, it cannot show how loads can influence the flange and web thickness; the web and flange thickness stay consistent throughout the model.

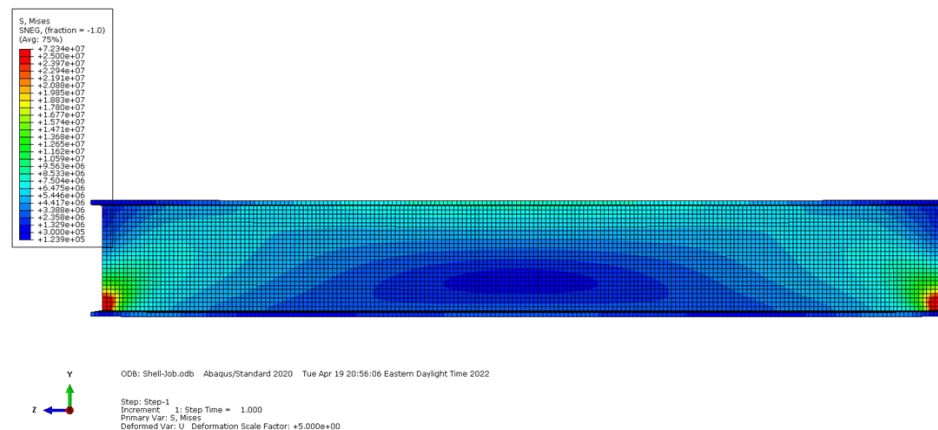


Figure 6: Shell Element

Finally, we took multiple days to analyze all the data retrieved from the solid element type to determine the difference in the solid elements relative to the other 2 element types. First, we noticed a difference in the order of magnitude between our von mises stress of the shell and solid beam elements, however, this discrepancy was less than 10% between the Shell and beam element mesh types, so we assumed we had made a fatal error. So, we decided to recreate the file again to see if this error could be remedied. After recreating it, the difference was still there, so we surmised that it must be a difference between element types that was causing this discrepancy. We also noticed an increase in the order of magnitude on max displacement, which was intriguing. We hypothesized that this could be the reason that our max stress differed by an order of magnitude between element types.

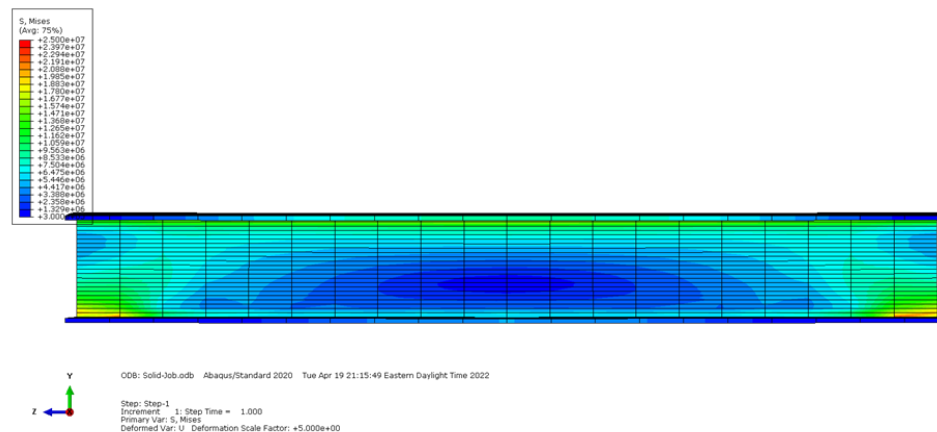


Figure 7: Solid Element

Conclusion:

After further analysis, we can confidently say that the solid element is the most accurate element type for beam simulations. We tried applying an eccentric load to each beam with differing results, but the solid element type seemed to emulate what we had expected. We would recommend the solid element type, if the computational expense is not a problem and if the user is well-versed in partitioning cells for meshing. The largest downside for this element type is the high computational expense as stated multiple times before due to mesh density. For the shell element type the number of elements increases as a function of $(mesh\ density)^2$, however for the solid element type the number of elements increases as function of $(mesh\ density)^3$. The mesh density had the most dramatic effect on the shell and solid beam elements, as we can see that the stress increases on an order of magnitude as the mesh size decreases. Most of the stress was focused on the point where the web and bottom flange meet, and this stress increased as the mesh density increased. If needed, increasing mesh density is an excellent way to determine where localized stress will be located on a beam.